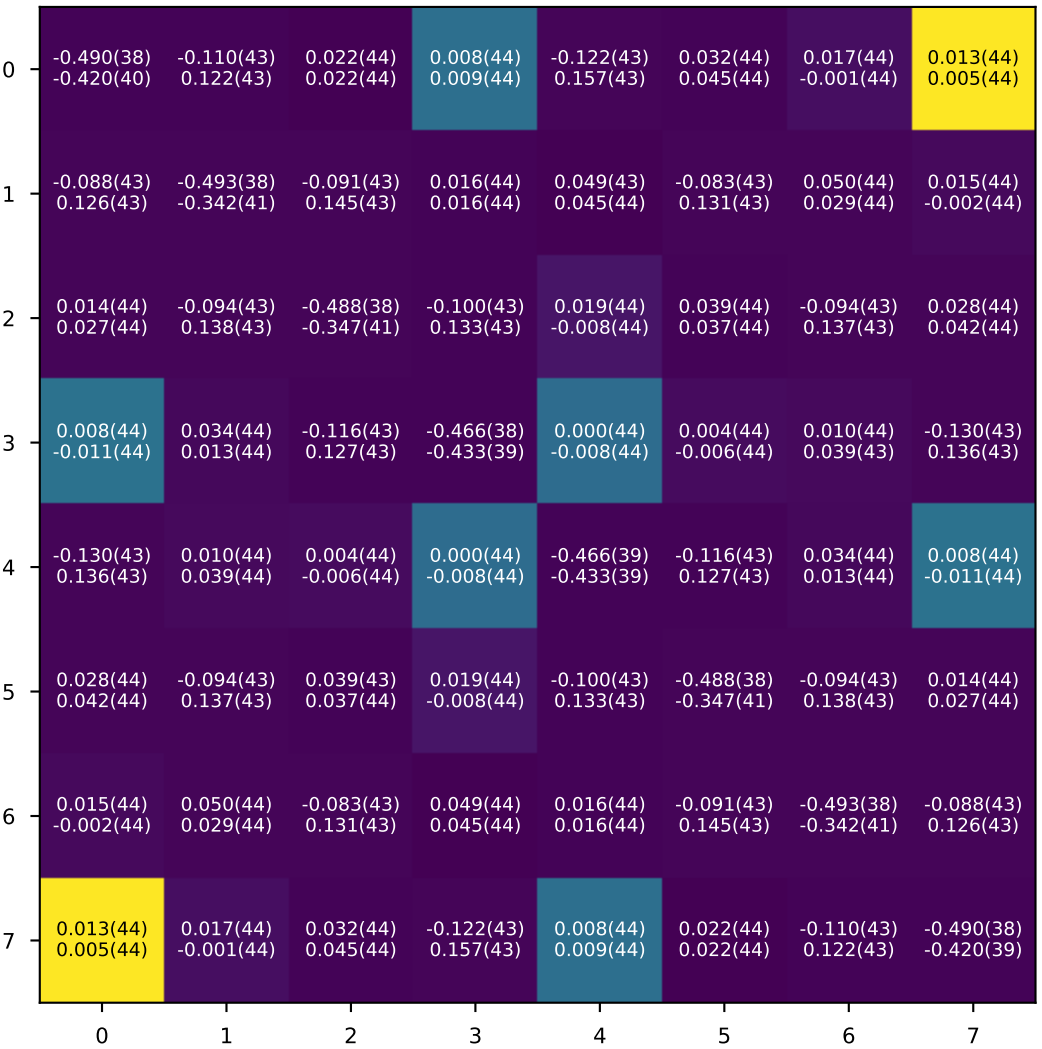
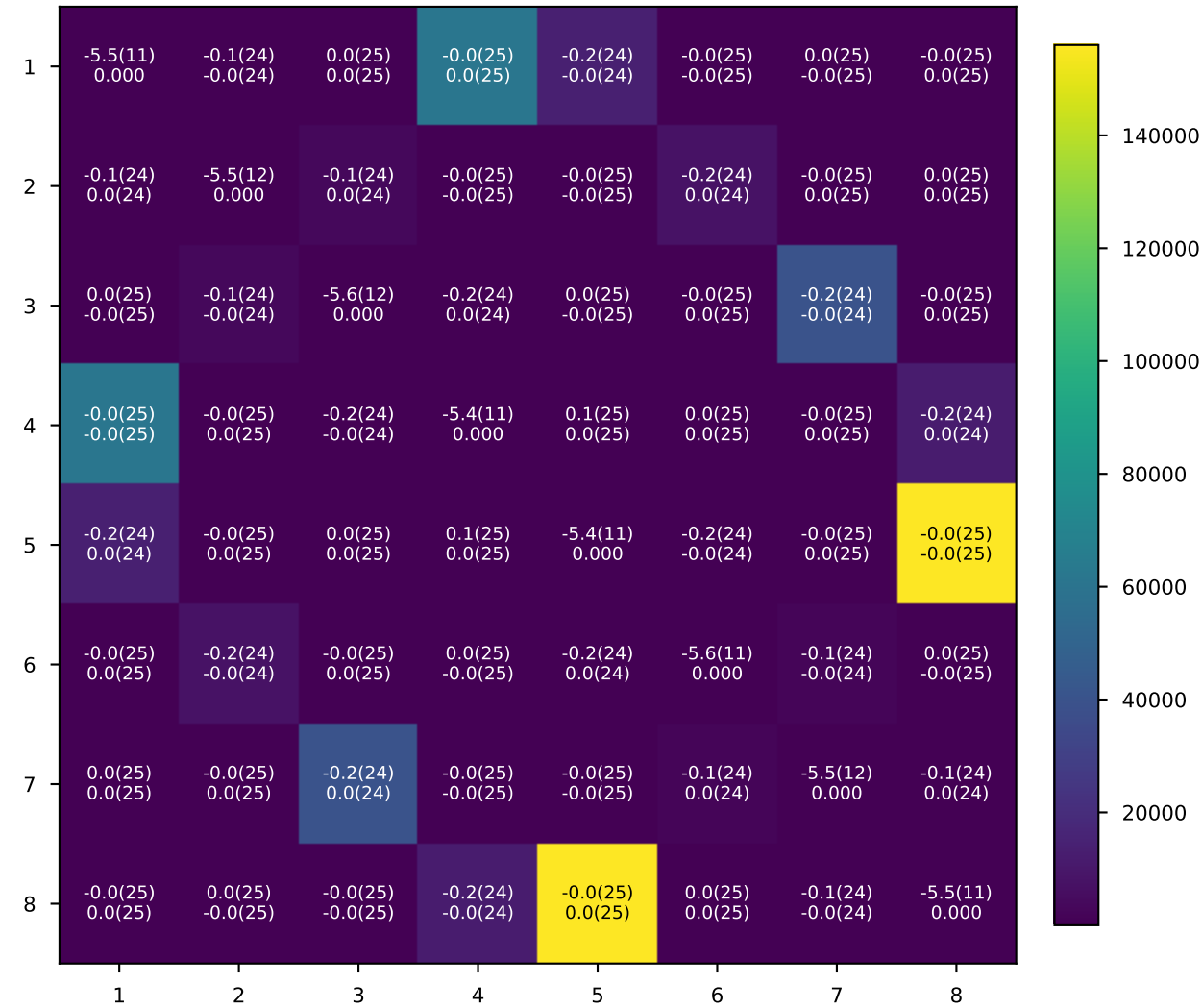
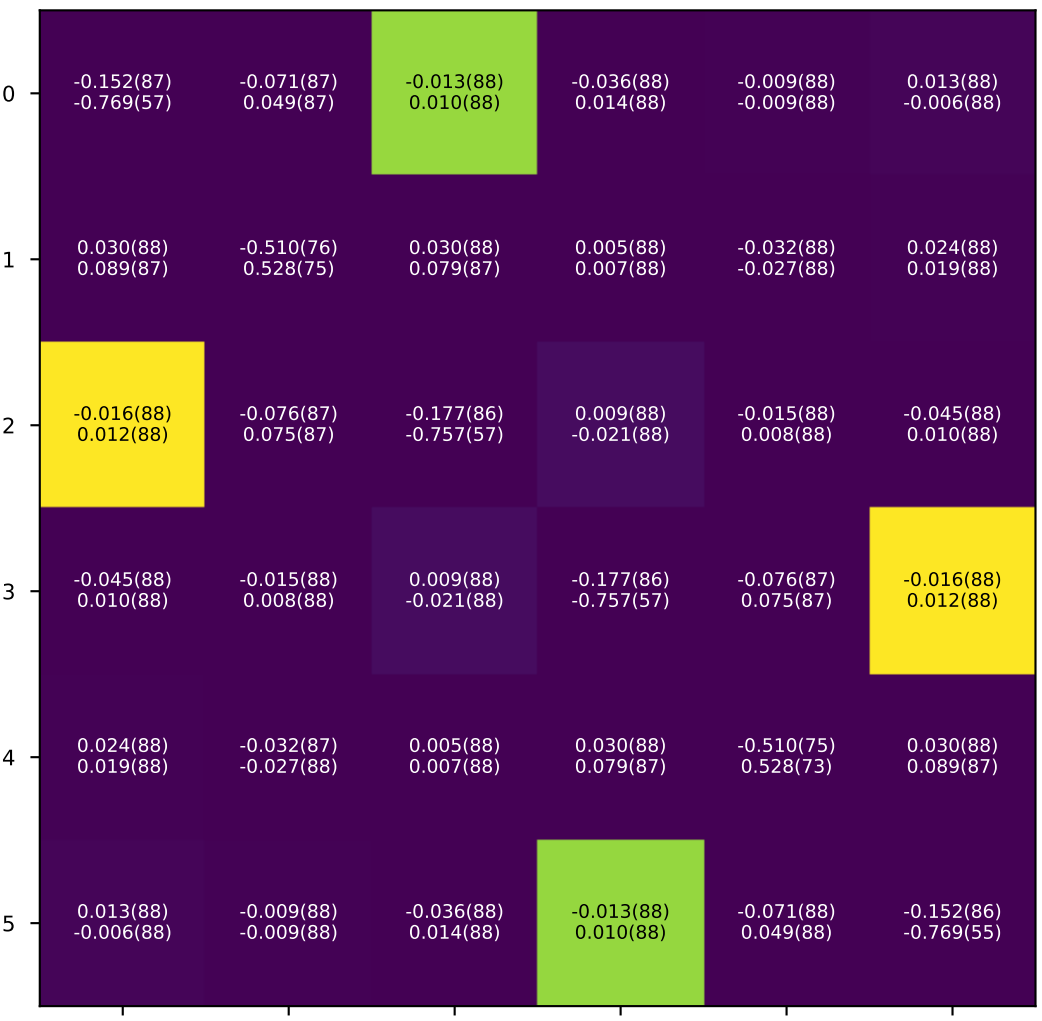
4 qubits - $\langle j|U|k\rangle$



4 qubits - $\langle j|U^\dagger HU|k\rangle$



3 qubits - $\langle j|U|k\rangle$



3 qubits - $\langle j|U^\dagger HU|k\rangle$